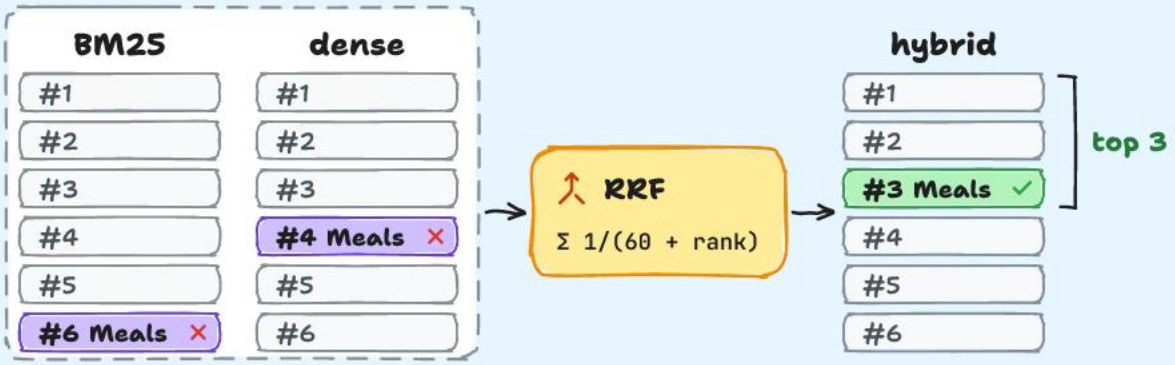
Two rankings, one list

Day 08: hybrid search, measured on Day 07's 81 labelled queries.

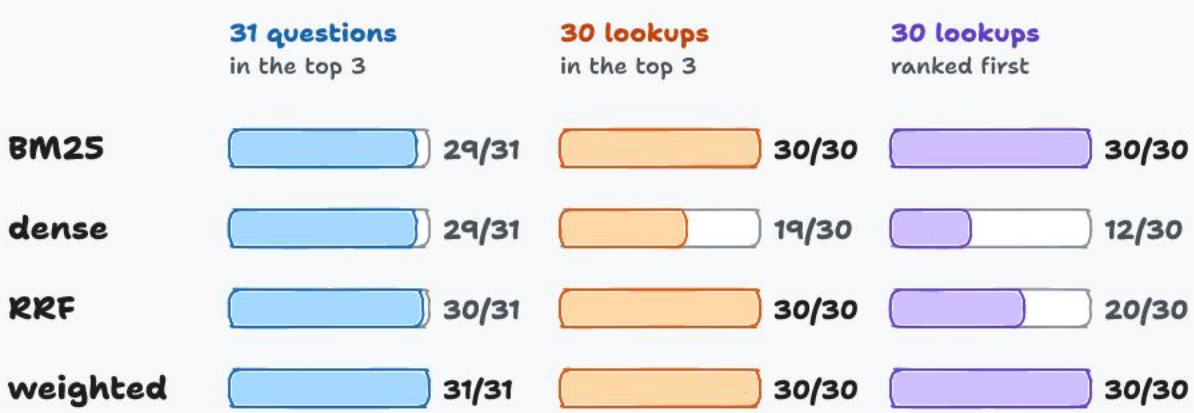
THE QUESTION BOTH RETRIEVERS MISSED

"Will Acme pay for wine with a business dinner?"



#6 and #4: both fairly high. Agreement lifts it to #3: 1/66 + 1/64.

THE SAME 81 QUERIES, 4 WAYS



RRF: no weight to tune, but 10 lookups lose first place

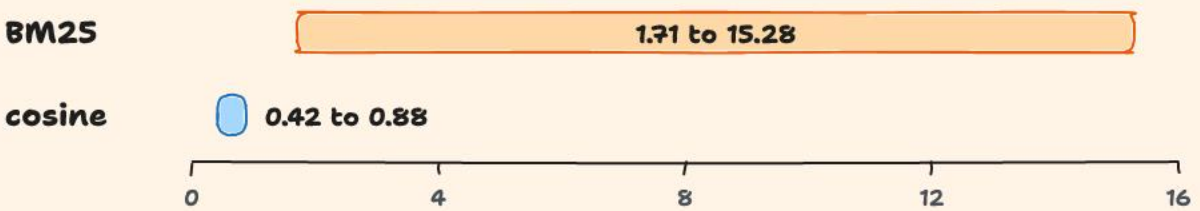
weighted: min-max scores, 0.7 dense + 0.3 BM25, weight chosen on dev

Fuse positions, not scores

Two scales that never meet, and a fix that uses ranks only.

WHY NOT ADD THE SCORES?

best score per query



added together, BM25 wins: the raw sum = BM25's top 3 for 62 of 71 queries

Σ RECIPROCAL RANK FUSION

```
score(chunk) = Σ 1 / (60 + its rank in each list)
```

`#1 → 0.0164` `#3 → 0.0159` `#10 → 0.0143` `not returned → 0`

chunk (wine question)	BM25	dense	hybrid	
expense › Scope	#1	#1	#1	
travel › Purpose	#2	#2	#2	
expense › Meals	#6	#4	**#3**	✓
vpn › installing	#9	#5	#4	

the answer: neither retriever's first, both fairly high

TIES ARE COMMON

(2, 5) and (5, 2) score the same: break by chunk id

k BARELY MATTERS HERE

k = 20 to 1000: identical counts on 81 queries

What RRF throws away

Both losses come from ignoring the scores.

1 AGREEMENT BEATS A SINGLE FIND

"How much time off do new dads get?"



BM25's 12 chunks: 2 votes each

none about parental leave

dense's top 3: 1 vote

the answer ends up #15

2 HOW SURE A RETRIEVER IS GETS LOST

lookup: "15 minutes"



dense: #19 ↓ RRF: #2

a lead of 2x becomes just "rank 1": 10 of 30 lookups lose first place

⚖ KEEP THE SCORES: A WEIGHTED FUSION

rescale each list to 0..1, then 0.7 × dense + 0.3 × BM25

the weight: chosen by a rule on 46 dev queries (not pre-registered)

31/31 questions in the top 3

30/30 lookups ranked first

15 test questions: RRF 15/15, weighted 15/15 in the top 3

a tie, and only a check: test results were seen while exploring

Before you ship hybrid

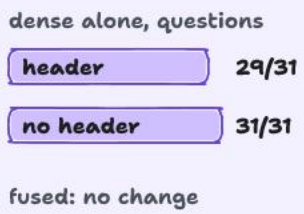
Three checks and a decision, from today's measurements.

SAME TEXT FOR BOTH RETRIEVERS?

alone, dense prefers no header

(2 questions, p = 0.5); fused, no gain

keep one text, one pipeline



FILTER EVERY RETRIEVER

shared BM25 index: a managers-only

chunk in 8 of alice's rankings

ranking low is still a leak



THE SECOND RETRIEVER IS CHEAP

embedding the question: ~10 ms

BM25 + fusion: < 0.1 ms (44 chunks)

the model call dominates



DECISION FOR THIS CORPUS

weighted fusion (0.7 dense + 0.3 BM25): no miss in the top 3

RRF kept as the no-labels alternative: nothing to tune

never add raw scores from different retrievers

next: Day 09 tests a reranker on the same 81 queries